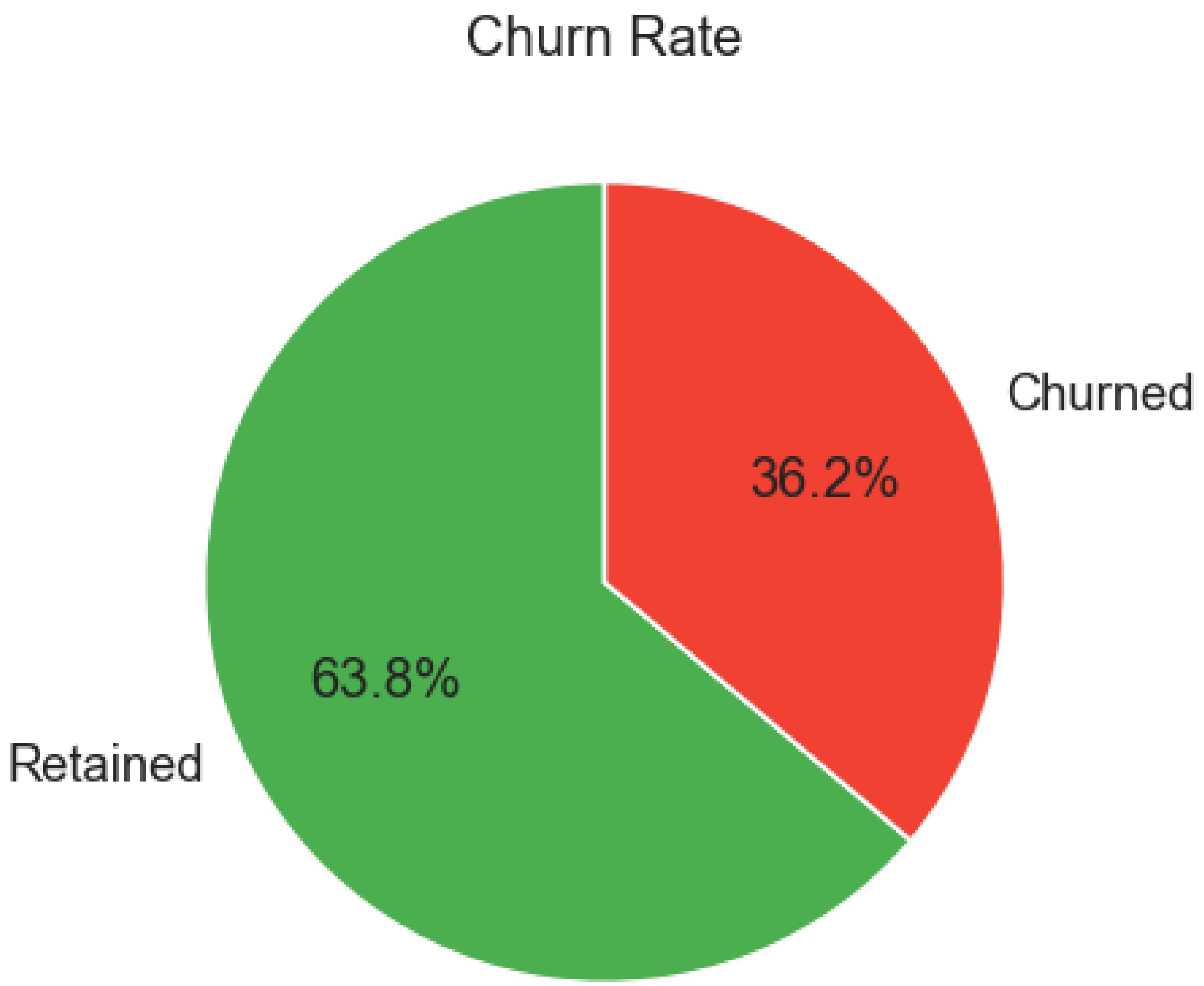
THE PROBLEM

A mid-sized telecom is losing customers 53% faster than the industry average.

2.3% monthly churn vs 1.5% industry benchmark. 2,531 customers churned out of 7,000.

MONTHLY CHURN	CUSTOMERS LOST	INDUSTRY BENCHMARK
2.3% +53% vs benchmark	**2,531** of 7,000	**1.5%** monthly

Churn Rate Distribution 36.2% / 63.8%



METHOD

EDA to find the story. XGBoost to predict it. SHAP to explain it. KMeans to act on it.

01 Exploratory analysis to identify raw churn signals before touching a model

02 XGBoost classifier trained on 5,600 customers, tested on 1,400 unseen

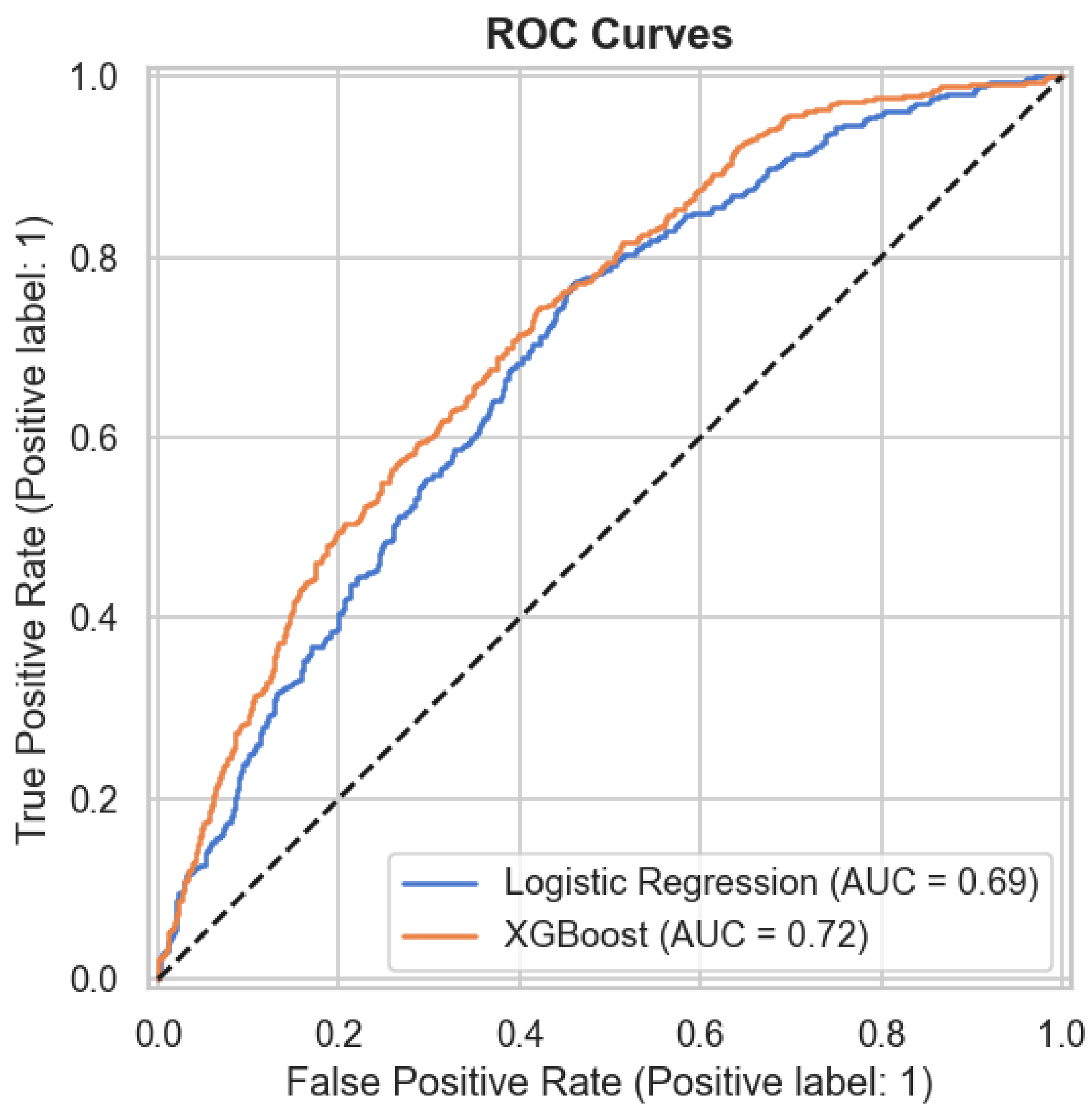
03 SHAP to explain every individual prediction, not just the model overall

04 KMeans (k=3) to group 5,505 predicted churners into actionable segments

05 Cost-aware threshold tuning, flagging the right customers beats raw accuracy

Model Comparison · ROC

AUC 0.7226 vs 0.6896



XGBOOST AUC
0.7226

LOGISTIC REG.
0.6896

LIVE DASHBOARD

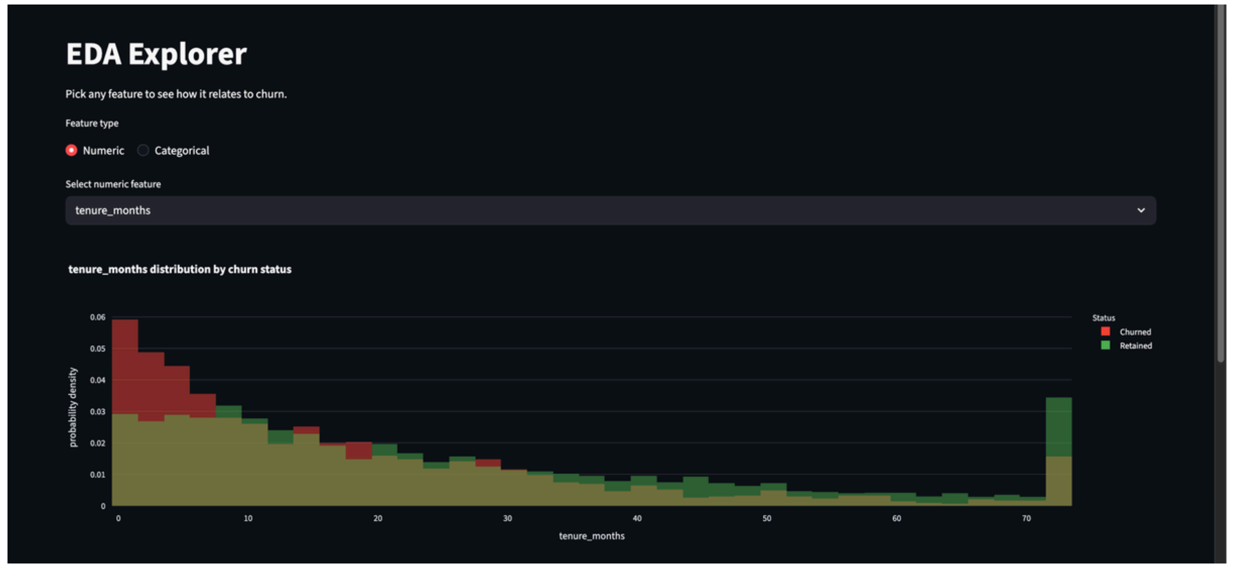
A four-page dashboard the CMO can hand to any retention analyst.

Hosted on HuggingFace Spaces · Built with Streamlit

Overview

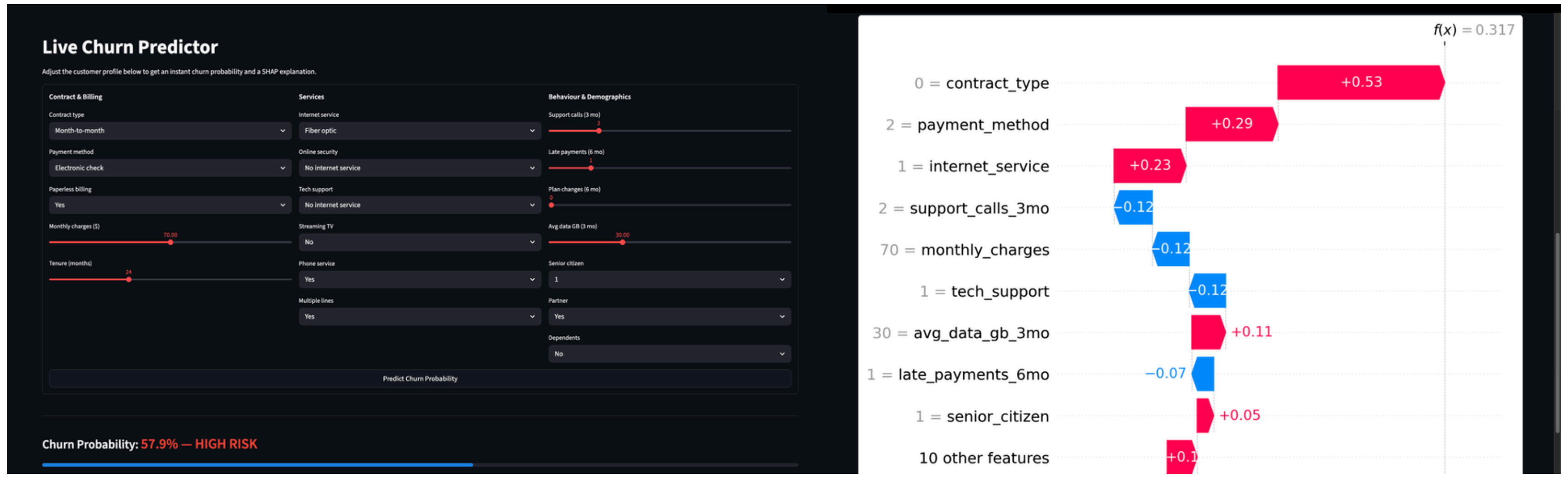
EDA Explorer

contract_type bar chart



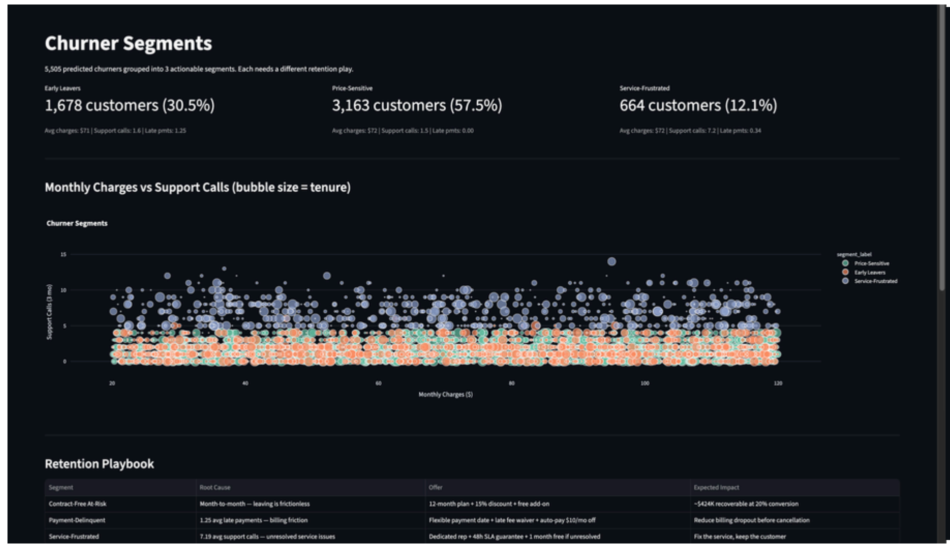
Predict

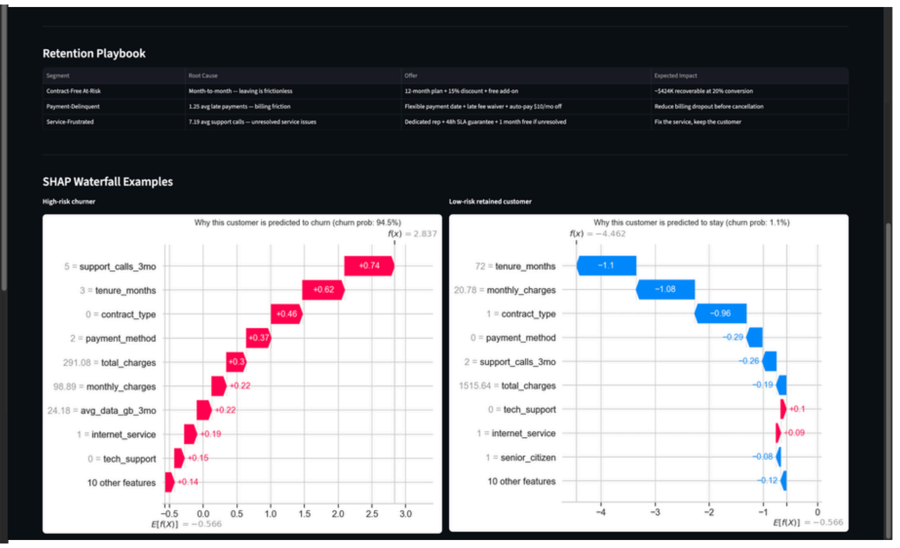
High-risk profile · SHAP waterfall



→SHAP waterfall explains WHY, not just what

Segments





RESULTS

0.7226 ROC-AUC. $213K recoverable revenue. Three segments, one offer each.

XGBOOST ROC-AUC
0.7226 vs 0.6896

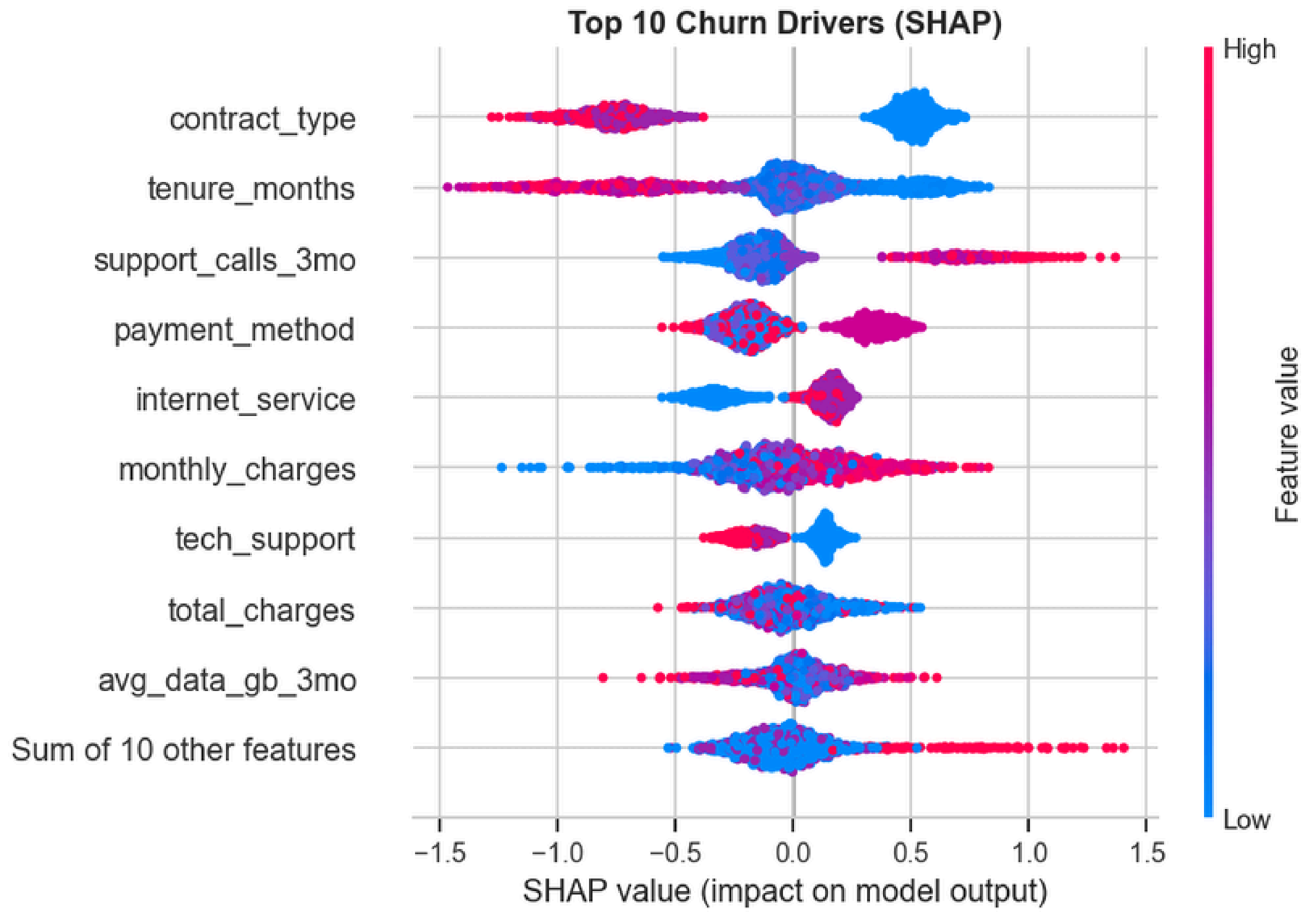
OPTIMAL THRESHOLD
0.15 cost-aware

REVENUE SAVED
$213,280 at θ=0.15

OFFER ECONOMICS
$100 → $670 cost · LTV

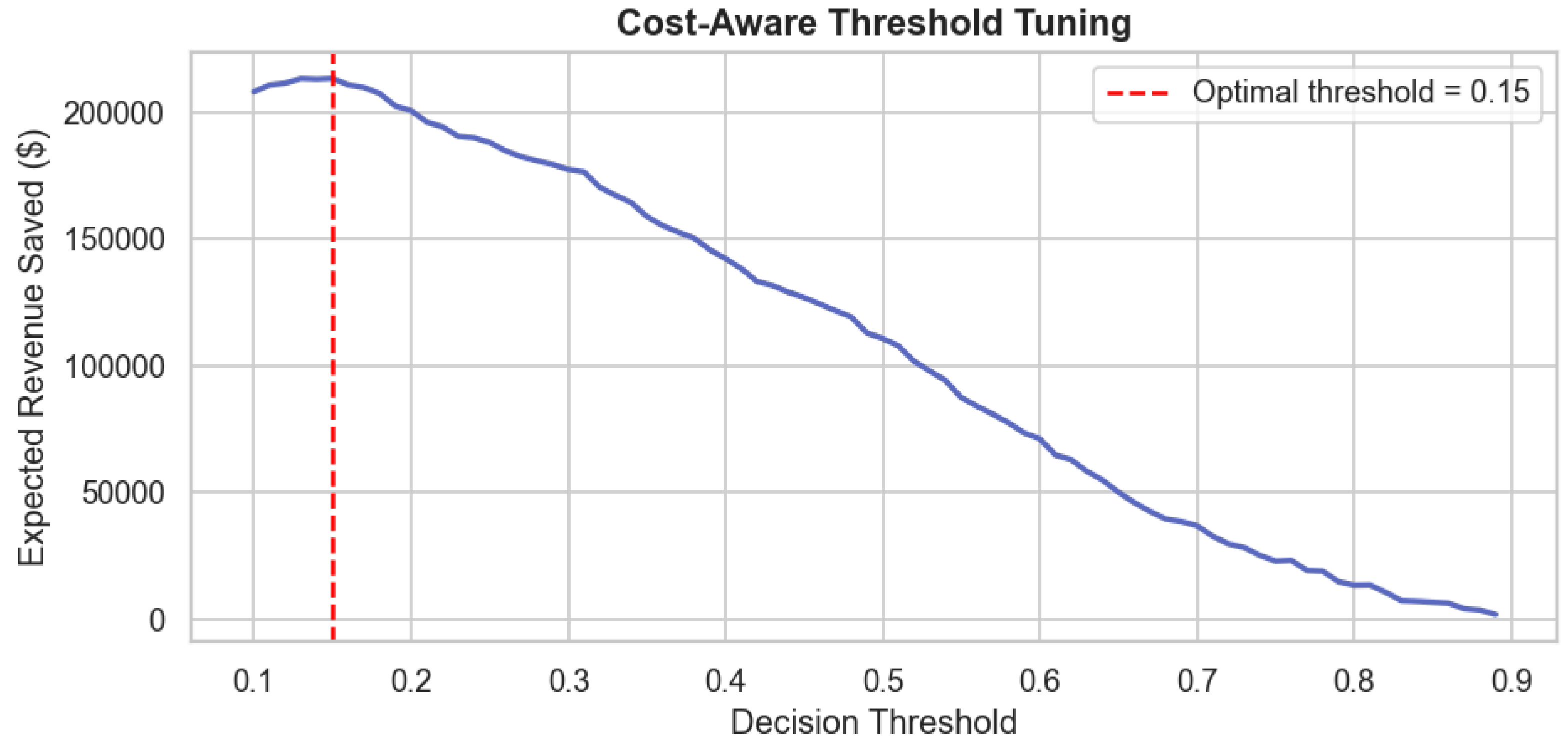
Top Global Drivers · SHAP

contract · tenure · charges · calls



Threshold vs Expected Revenue

optimal θ = 0.15



CONTRACT-FREE AT-RISK
~$424K
recoverable @ 20% conversion
3,163 customers

SERVICE-FRUSTRATED
7.19
avg support calls / 3 mo
664 customers

PRODUCTION ROADMAP

The model works. Now make it production-grade.

Five concrete bets for the next sprint, ranked by impact on retention revenue and model durability.

01 **Hyperparameter search on XGBoost**
Expect 2–3 more AUC points with Optuna. Low effort, high return.

02 **A/B test framework**
50% of predicted churners get the offer, 50% holdout. Measure churn at Day 30 and Day 60.

03 **Data drift monitor**
Weekly PSI on contract_type and monthly_charges so the model doesn't go stale after a pricing change.

04 **Live feature ingestion**
Replace the static CSV with a live pull from the billing warehouse.

05 **Role-based dashboard access**
Retention analysts see the offer recommendation, not the raw SHAP values.